

The Labor Demand and Labor Supply Channels of Monetary Policy

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BGGMP Reading Group, NYU Economics

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LABOR SUPPLY CHANNEL OF MONETARY POLICY?

*“Policies to support labor supply are not the domain of the Fed;
Our tools work principally on demand”*

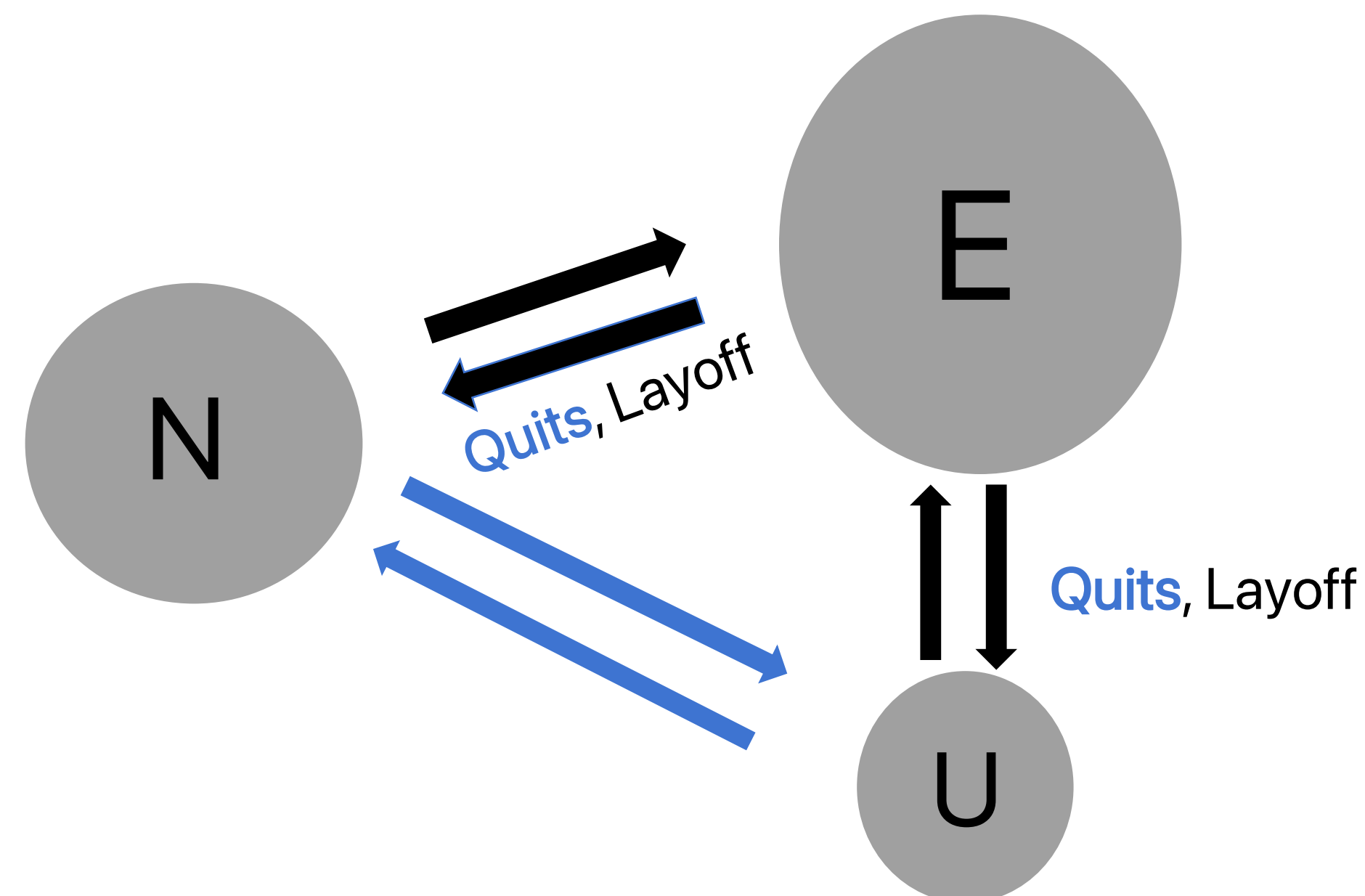
- Jerome Powell, Nov 30th, 2022

- General Consensus on Transmission of MP shock : Direct effect to L^d , less focus on L^s
(e.g. : IS-LM Framework, NK Model with Sticky Wage)
- This paper: There is a substantial labor supply response (= income effect) to MP shock
- The labor supply channel halves the decline in employment due to contractionary MP shock

LABOR SUPPLY CHANNEL OF MONETARY POLICY!

Challenge: How to identify the labor supply response?

- Identify labor market flows (plausibly) driven by labor supply considerations:
 1. $U \rightarrow N$: You decide to stop active search
 $N \rightarrow U$: You decide to start active search
 2. $E(\text{Quits}) \rightarrow N, U$: You decide to quit

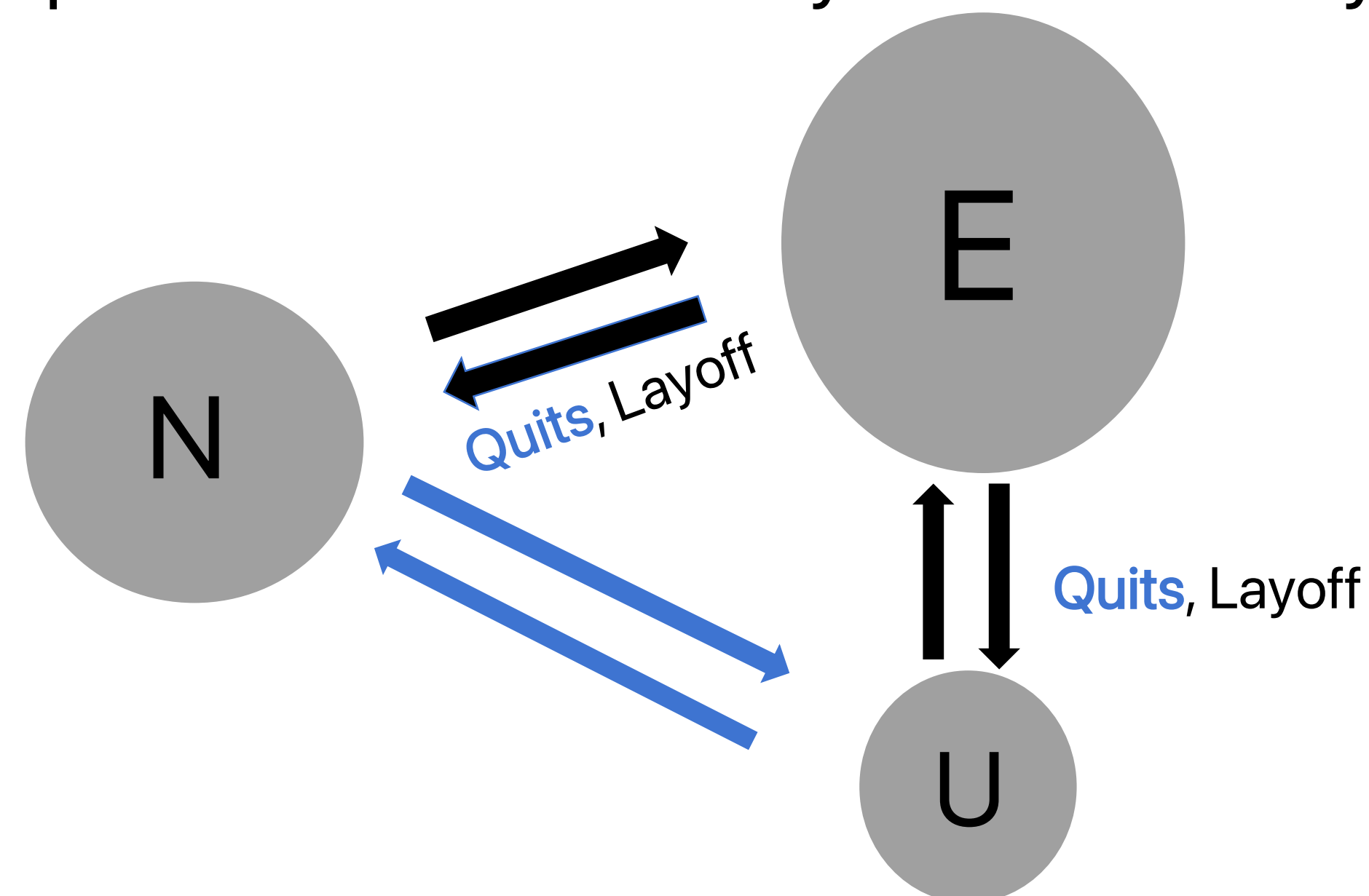


Blue: Affected by L^s

LABOR SUPPLY CHANNEL OF MONETARY POLICY!

This Paper: Uses Monthly CPS Data to document:

- Work-initiated quits comprise large portion of outflow from E (40% of EN flow)
- Upon contractionary MP shock, significant labor supply response
 - More job search: UN flow decreases and NU flow increases
 - Less Quits: Quits components of EN, EU flow decrease
- Labor supply response dampens the contractionary effect of MP by half



Blue: Affected by L^s

ROADMAP

- Document two facts about the labor flows:
 - 1) $N \leftrightarrow U$ flows are large and reflect counter-cyclical job searching behavior
 - 2) Work-initiated quits comprise large portion of outflow from E (40% of EN flow)
- Show significant labor supply response to contractionary MP shock
- Build a sticky-wage model where labor supply decisions are important

STYLIZED FACTS ON LABOR FLOWS

- Data: Monthly CPS 1978 — 2019, Seasonally Adjusted

[Table 1 : Cyclicalities of Labor Stocks]

	Employment- Population Ratio	Unemployment	Labor Force Participation
$\text{mean}(x)$	61.14	6.19	65.16
$\text{std}(x)/\text{std}(Y)$	0.72	8.25	0.23
$\text{corr}(x, Y)$	0.83	-0.85	0.35

Labor State E:
Less volatile, pro cyclical

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Labor State U:
Very volatile, Counter-cyclical

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Labor Force Participation:
Not volatile, modestly pro-cyclical

STYLIZED FACTS ON LABOR FLOWS

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- Flows across Labor States: Calculate the Markov Transition Prob.

$$\begin{bmatrix} E \\ U \\ N \end{bmatrix}_{t+1} = \begin{bmatrix} 1 - p_{EU} - p_{EN} & p_{UE} & p_{NE} \\ p_{EU} & 1 - p_{UE} - p_{UN} & p_{NU} \\ p_{EN} & p_{UN} & 1 - p_{NE} - p_{NU} \end{bmatrix}_{t+1} \begin{bmatrix} E \\ U \\ N \end{bmatrix}_t.$$

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- [Table 2] : Labor Flows Dynamics

<i>From</i>	<i>To</i>				p_{EU}	p_{EN}	p_{UE}	p_{UN}	p_{NE}	p_{NU}
	<i>E</i>	<i>U</i>	<i>N</i>							
<i>E</i>	0.956	0.014	0.030	mean	0.014	0.030	0.255	0.226	0.046	0.025
<i>U</i>	0.255	0.519	0.226	std(x)/std(Y)	5.19	2.46	5.69	4.14	3.00	5.22
<i>N</i>	0.046	0.025	0.929	corr(x , Y)	-0.83	0.49	0.78	0.71	0.65	-0.68

LABOR SUPPLY CHANNELS: OVERVIEW

	p_{EU}	p_{EN}	p_{UE}	p_{UN}	p_{NE}	p_{NU}
mean	0.014	0.030	0.255	0.226	0.046	0.025
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Labor Supply Channel 1

- Pro-cyclicality of U to N : In recession, less stop searching
- Counter-cyclicality of N to U: In recession, more actively search

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Labor Supply Channel 1

- Pro-cyclicality of U to N : In recession, less stop searching
- Counter-cyclicality of N to U: In recession, more actively search

Labor Supply Channel 2

- Less Quits from E to either U or N
- To see this, we need to decompose the EU, EN flows into layoffs / quits

DECOMPOSING INTO QUILS AND LAYOFFS

EU Flow Decomposition: CPS questionnaire asks reason for unemployment

- "Quit" if the reason is "job leaver",
- "Layoffs" if the reason is "job loser / layoff, job ended"
- "Others" if others ("re-entrant" or "new entrant". About 10-15%)

EN Flow Decomposition: Asks reason for leaving previous job (outgoing rotation group interviewed at 4th, 8th Month)

- Analyze those (1) not in labor force, (2) had worked within the past 12 months
- "Quit" if reason is "personal, family or school", "unsatisfactory work arrangements"
- "Layoffs" if reason is "slack work or business conditions"
- "Others" if others (usually retirees or disabilities. About 50%)

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- "Others" if others (usually retirees or disabilities. About 50%)

	EU Flows				EN Flows			
	Total	Quits	Layoffs	Other	Total	Quits	Layoffs	Other
mean	0.014	0.002	0.010	0.003	0.030	0.012	0.003	0.015
std(x)/std(Y)	5.19	8.11	7.39	5.44	2.46	5.88	14.42	4.80
corr(x , Y)	-0.83	0.60	-0.85	-0.30	0.49	0.53	-0.44	0.25

HOW IMPORTANT ARE QUITTS?

	Total	EU Flows			Total	EN Flows		
		Quits	Layoffs	Other		Quits	Layoffs	Other
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EU Flows:

- 1) 70% of EU flows due to layoffs: countercyclical
- 2) 10-15% of EU flows due to quits: pro-cyclical (less quits in recession)

HOW IMPORTANT ARE QUITTS?

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EN Flows: (Novel Facts)

- 1) EN Flows larger than EU Flows
- 2) 40% of EN flows due to quits: pro-cyclical (less quits in recession & quantitatively large)

- Flows between U and N, and quits are from labor supply decisions
- Evidence of significant income effect of L^s
 - Flows between U and N implies more job-search activities during recession
 - Quits are large and pro-cyclical.
- Next: Monetary Policy VARs using High-Frequency Identification
 - Contractionary MP shock leads to a significant increase in L^s via all above channels!
 - Mitigates the drop in employment due to weak labor demand

HIGH-FREQUENCY IDENTIFICATION OF MP SHOCK

High-freq. identification of MP shock aggregated to monthly frequencies

- Use narrow window to rule out endogeneity / causality problem (Gertler and Karadi (2015))
- Use Bauer and Swanson (2023): Include surprise around Fed Chair speeches & orthogonalized with respect to news BS2023

$$E[z_t \varepsilon_t^{mp}] \neq 0$$

Relevance Condition for Instrument

$$E[z_t \varepsilon_t^{-mp}] = 0$$

Exogeneity Condition

High-freq. surprise z_t instrument for ε_t^{mp}

IDENTIFICATION

HIGH-FREQUENCY IDENTIFICATION OF MP SHOCK

Baseline Six Variables SVAR-IV with 25bp MP shock to 2-yr Treasury Rate

- (1) 2-Yr Treasury Rate, (2) Unemployment Rate, (3) LF Participation Rate,
(4) Consumer Price Index, (5) Industrial Production, (6) GZ Excess Bond Premium

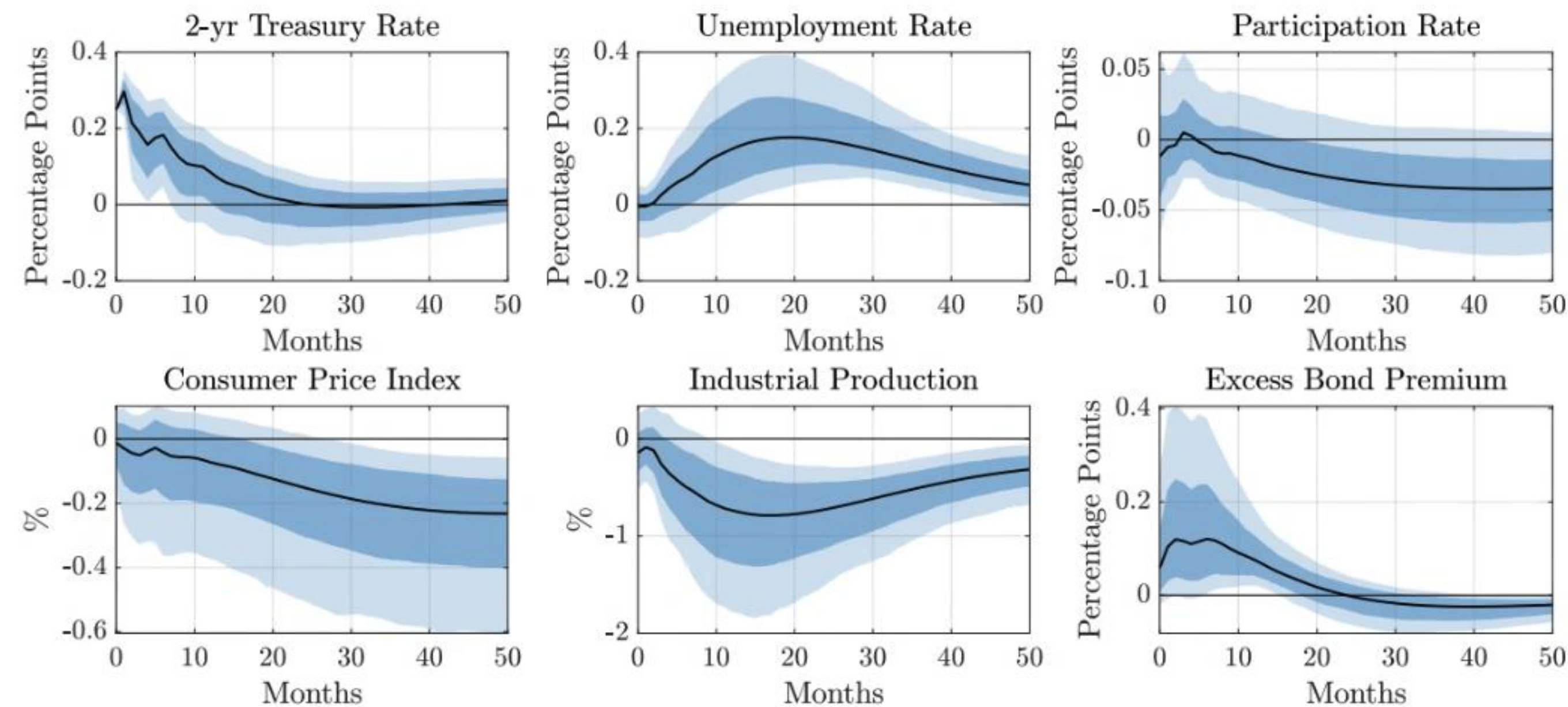
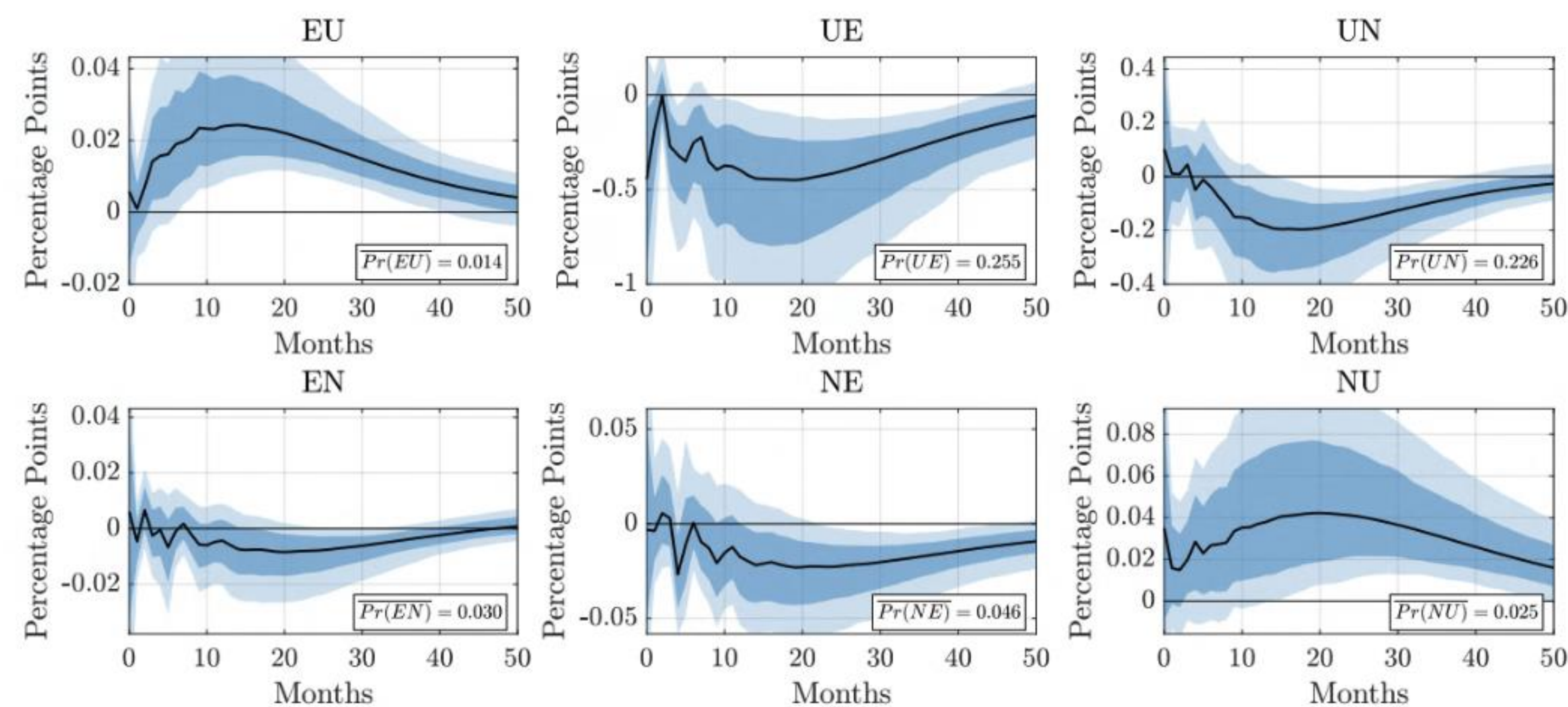


Figure 1 : Estimated IRF to 25bp MP Tightening (68% and 90% CI)

RESPONSE OF FLOWS TO A MP SHOCK

Extend the VAR to include 6 labor market flows one-by-one

FIGURE 2. Response of Labor Market Flows to a Monetary Policy Shock



OTHERS

Figure 1 : Estimated IRF to 25bp MP Tightening (68% and 90% CI)

RESPONSE OF FLOWS TO A MP SHOCK

- Increase in EU (driven by layoffs), decline in UE reflect weak labor demand

FIGURE 2. Response of Labor Market Flows to a Monetary Policy Shock

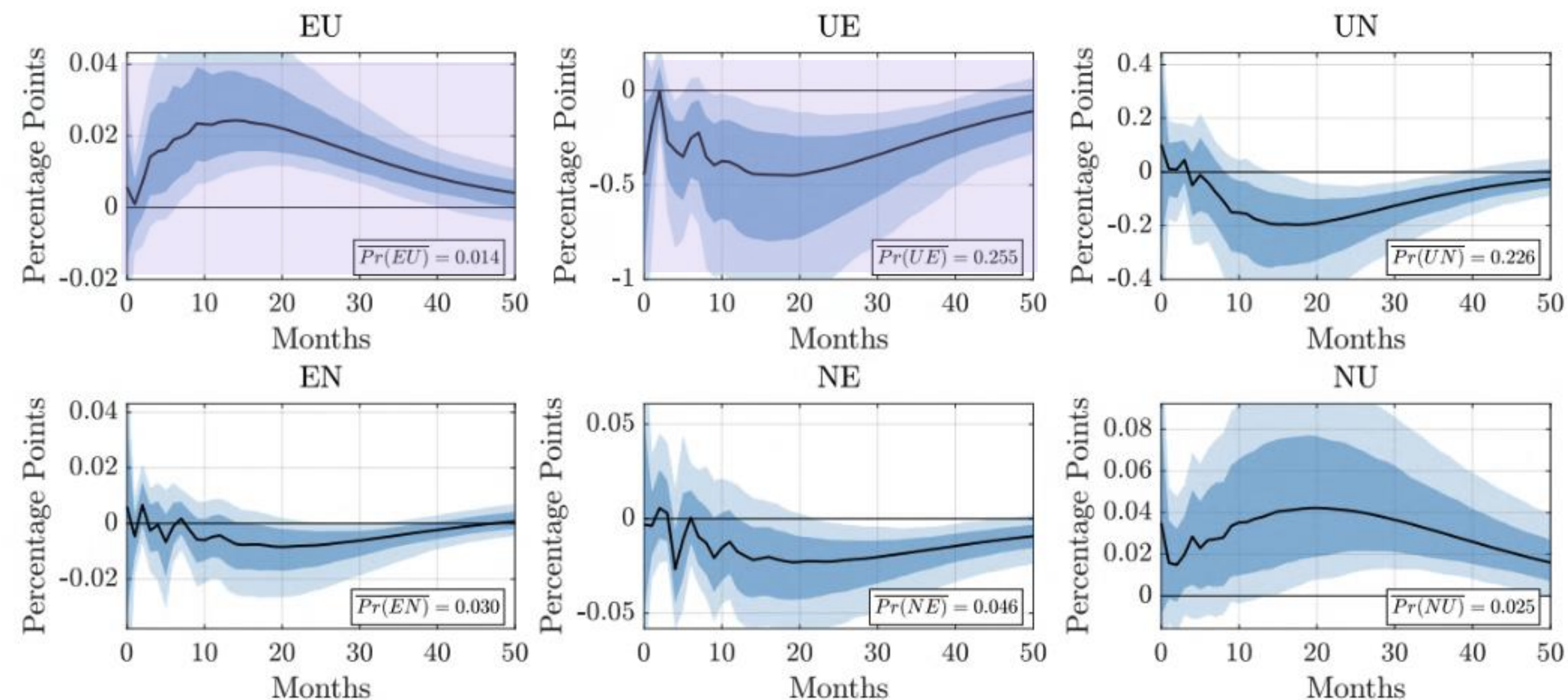


Figure 1 : Estimated IRF to 25bp MP Tightening (68% and 90% CI)

RESPONSE OF FLOWS TO A MP SHOCK

- Labor Supply Channel 1: UN decreases (less passive search), NU increases (more active search)

FIGURE 2. Response of Labor Market Flows to a Monetary Policy Shock

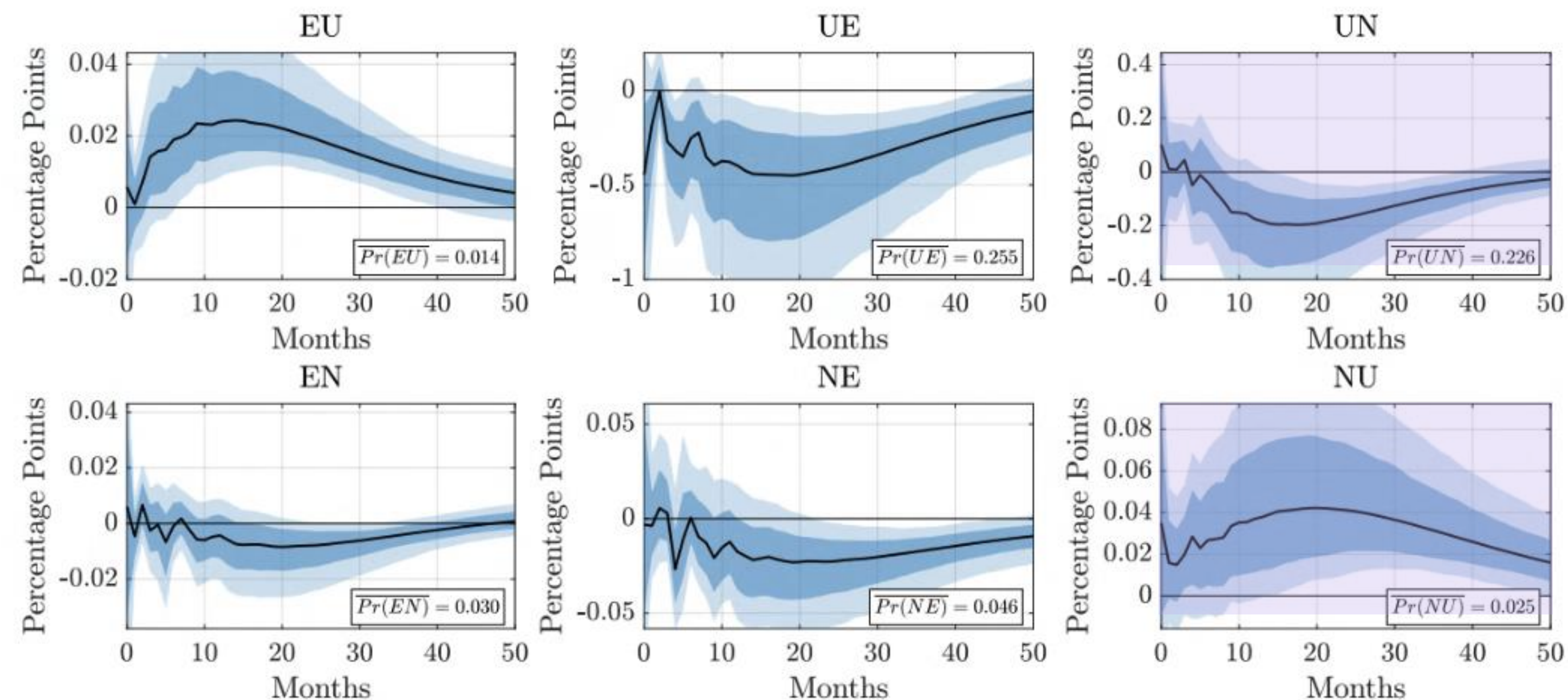


Figure 1 : Estimated IRF to 25bp MP Tightening (68% and 90% CI)

RESPONSE OF FLOWS TO A MP SHOCK

- Labor Supply Channel 2: Quits decline. Not obvious right away.
- Need to decompose EU, EN flow into quits and layoffs

FIGURE 2. Response of Labor Market Flows to a Monetary Policy Shock

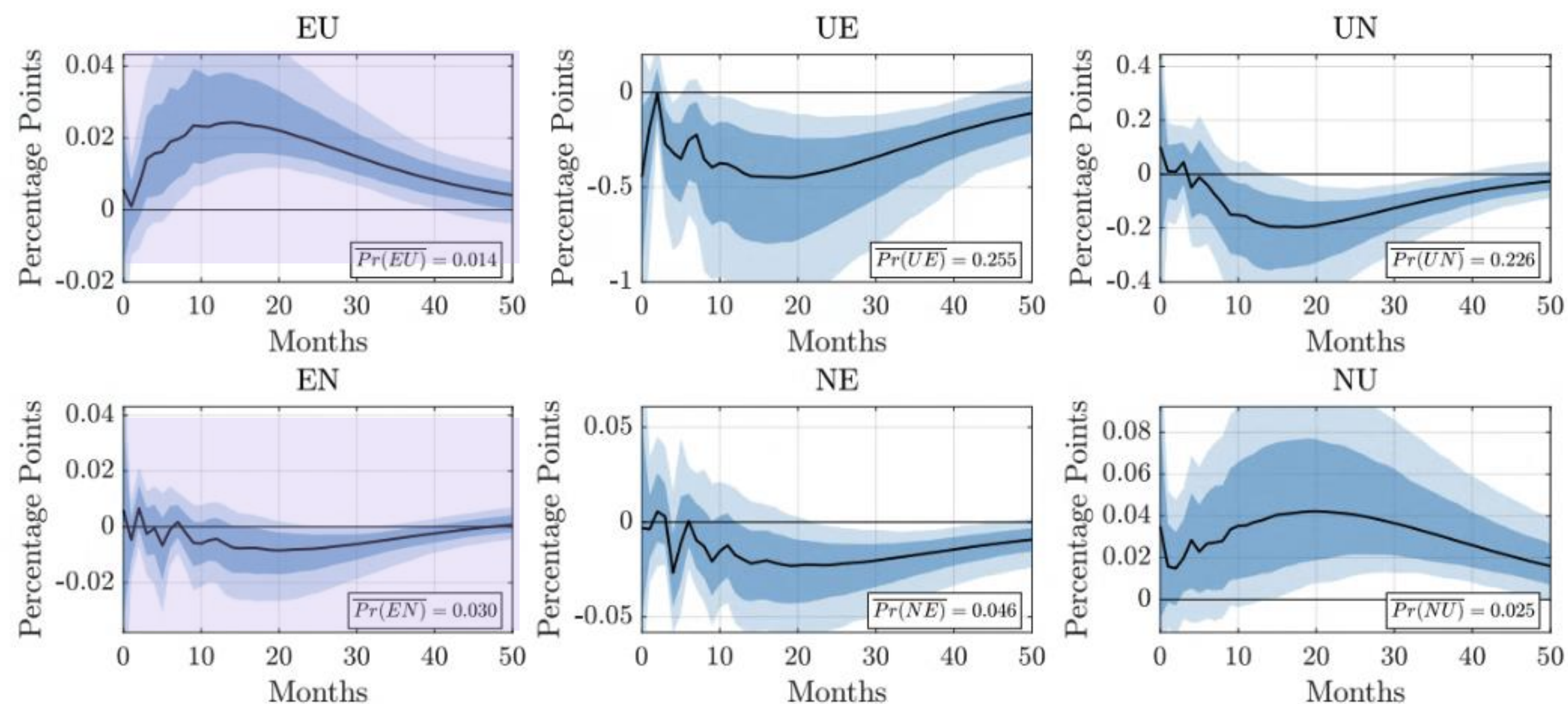
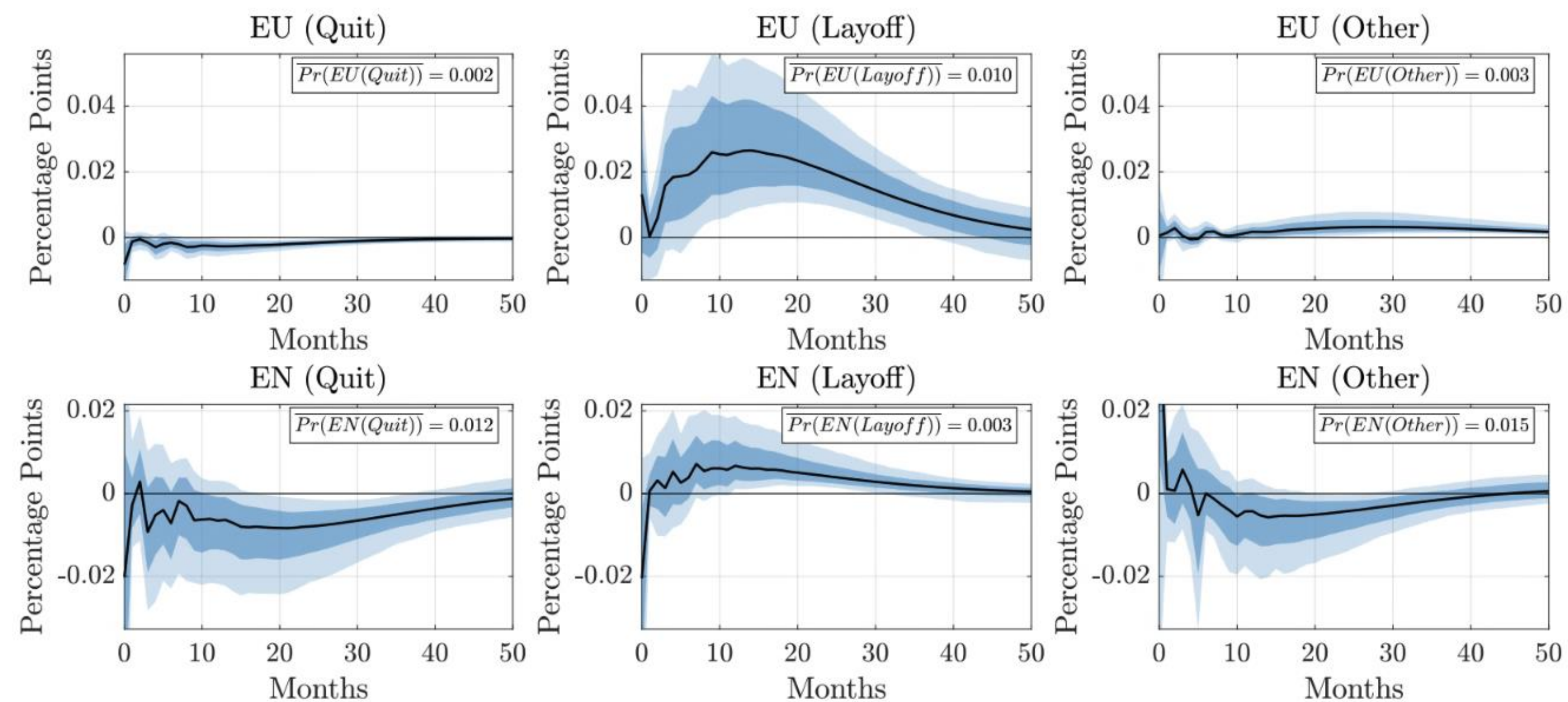


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DECOMPOSING EU AND EN RATES

Extend the VAR to include 6 labor market flows one-by-one

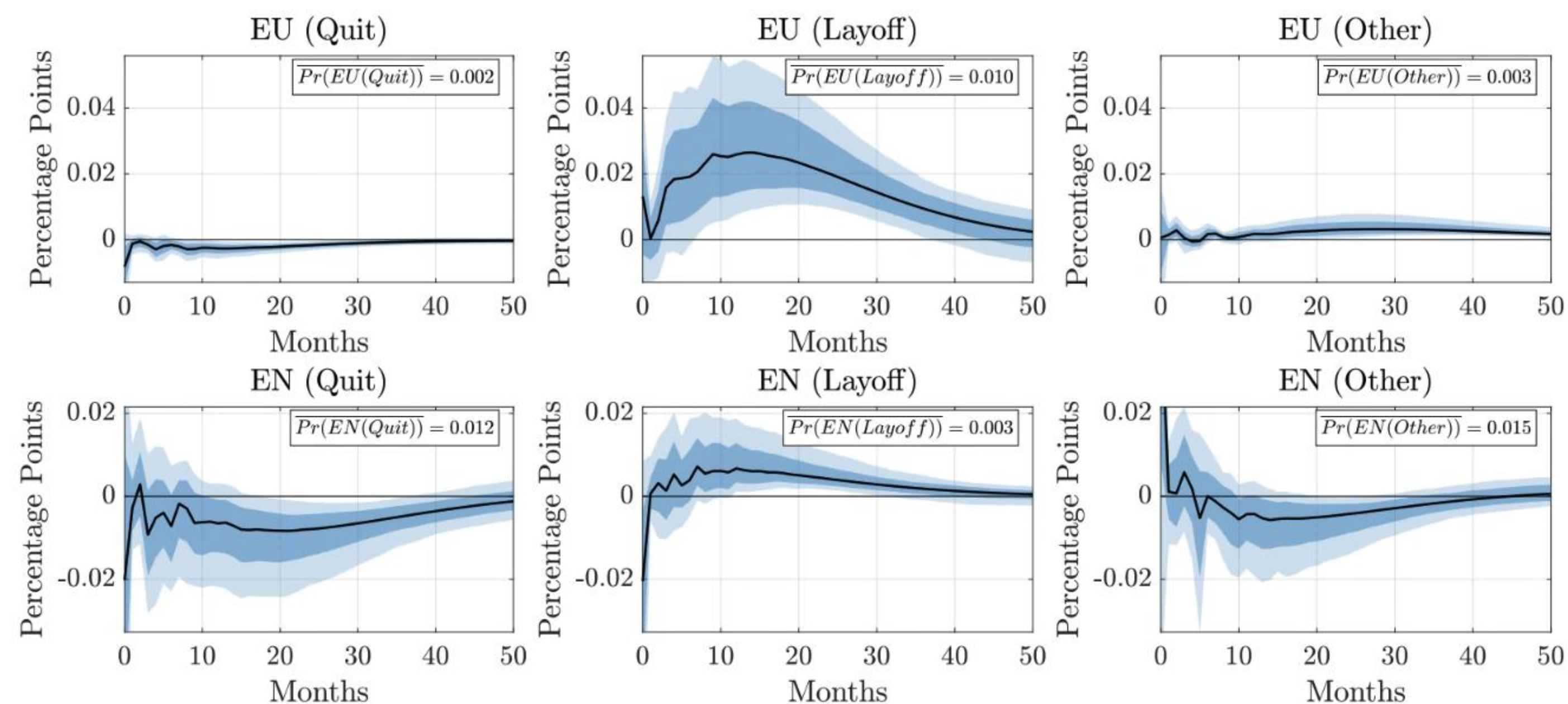
FIGURE 3. Decomposition of E-U and E-N Responses



DECOMPOSING EU AND EN RATES

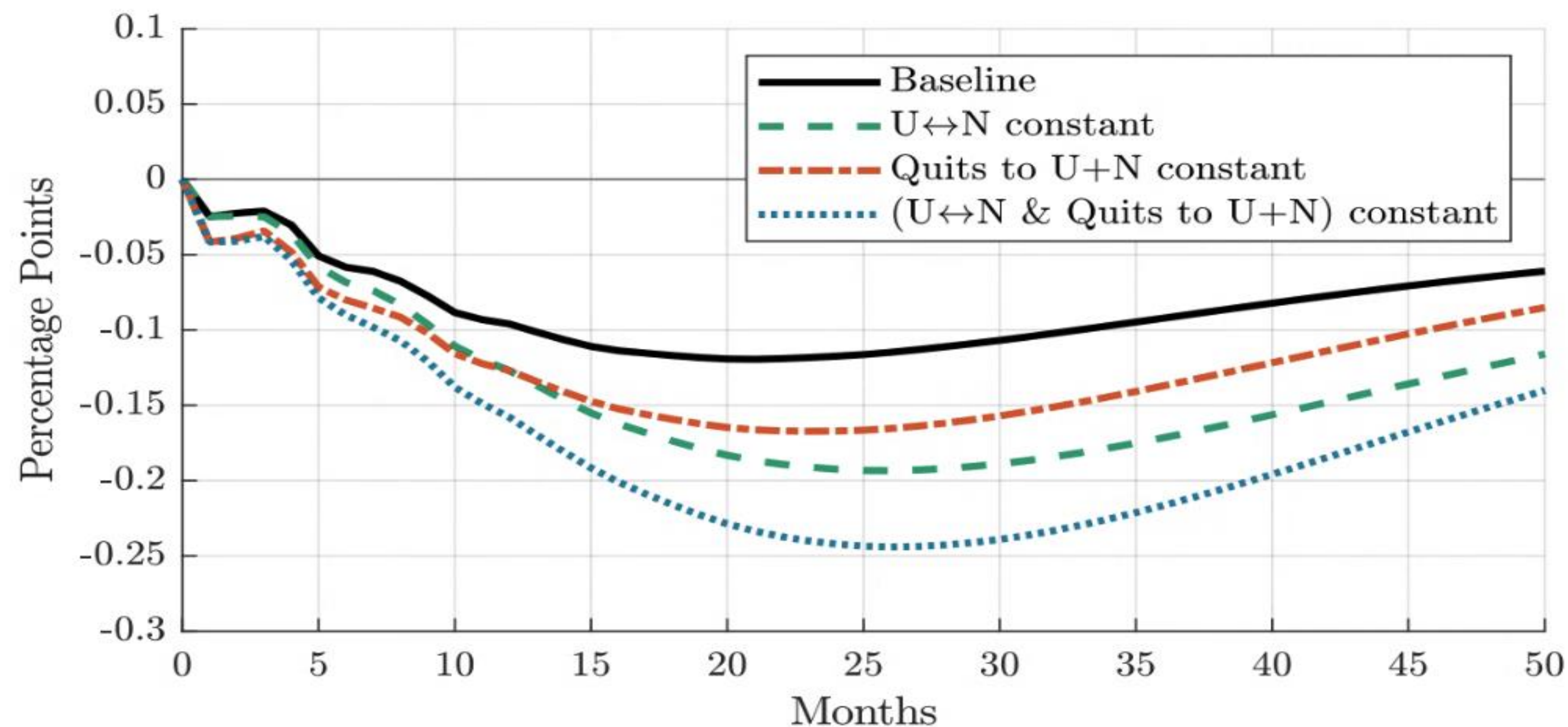
- Increase in layoff: Labor Demand Channel
- Quits decline: Labor Supply Channel
 - Large response of EN quits

FIGURE 3. Decomposition of E-U and E-N Responses



QUANTIFYING THE LABOR SUPPLY CHANNEL

- Figure: Response of Employment - Population Ratio to MP Shock
 - Counterfactual IRF constructed by shutting down each channel
- $U \leftrightarrow N$ flows more important (60%) than removal of less quits (40%)
- Without both channels, employment drop nearly doubles



Details

POTENTIAL ISSUE: COMPOSITION EFFECT

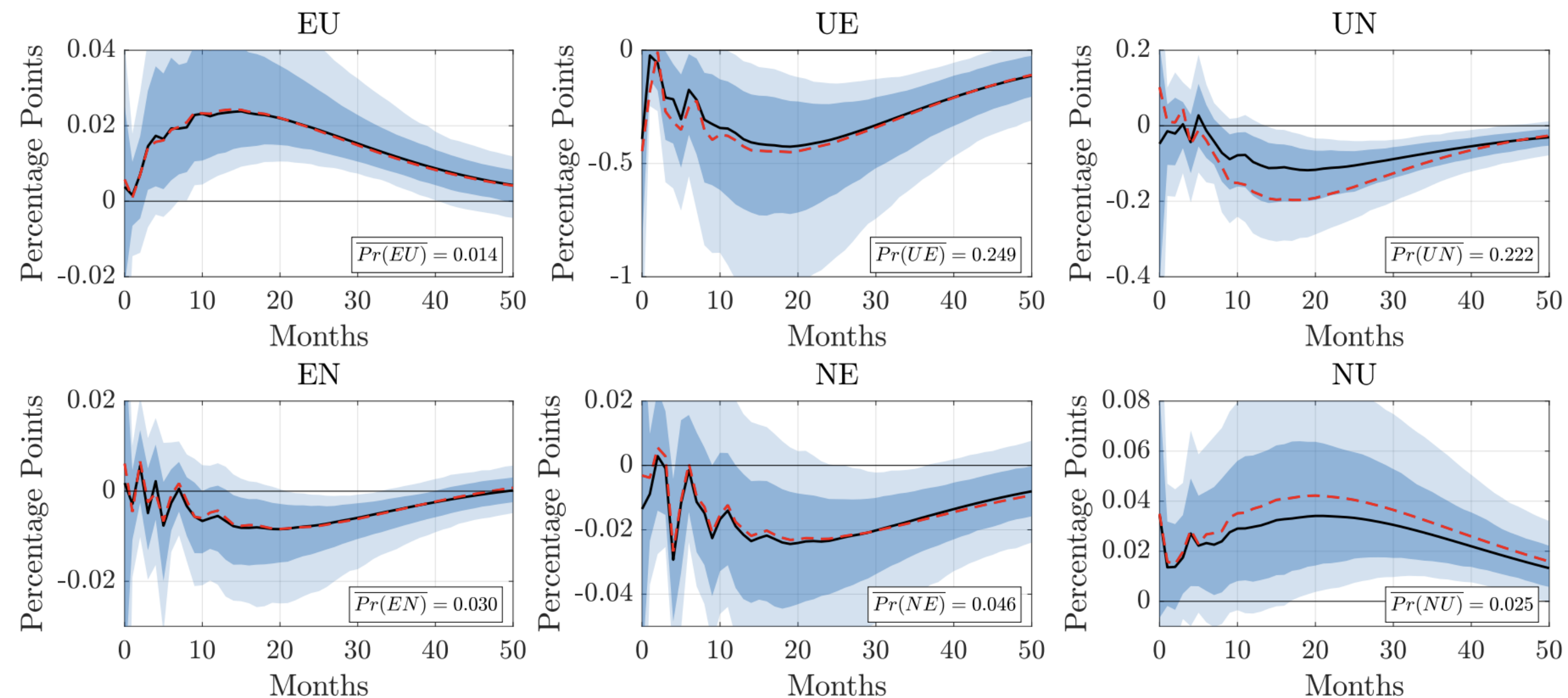
- Does the IRF truly reflect changes in individual behavior?
- Only capturing changes in group composition? (e.g.: pool of unemployed changes according to shocks)
- To reflect changes in individual behavior only, need to remove composition effect:

$$y_t - \bar{y} = \underbrace{\sum_i (y_{i,t} - \bar{y}_i) \cdot \bar{\omega}_i}_{\text{change in } y_{i,t}} + \underbrace{\sum_i \bar{y}_i \cdot (\omega_{i,t} - \bar{\omega}_i)}_{\text{change in } \omega_{i,t}} + \underbrace{\sum_i (y_{i,t} - \bar{y}_i)(\omega_{i,t} - \bar{\omega}_i)}_{\text{covariance}}.$$

- Consider 72 sub-groups for U & 24 for E and N (Age, gender, education, reason for unemployment, etc.)
- Construct composition-adjusted labor market flow & IRF again

POTENTIAL ISSUE: COMPOSITION EFFECT

Figure: Response of Composition-Adjusted Flows to MP Shock

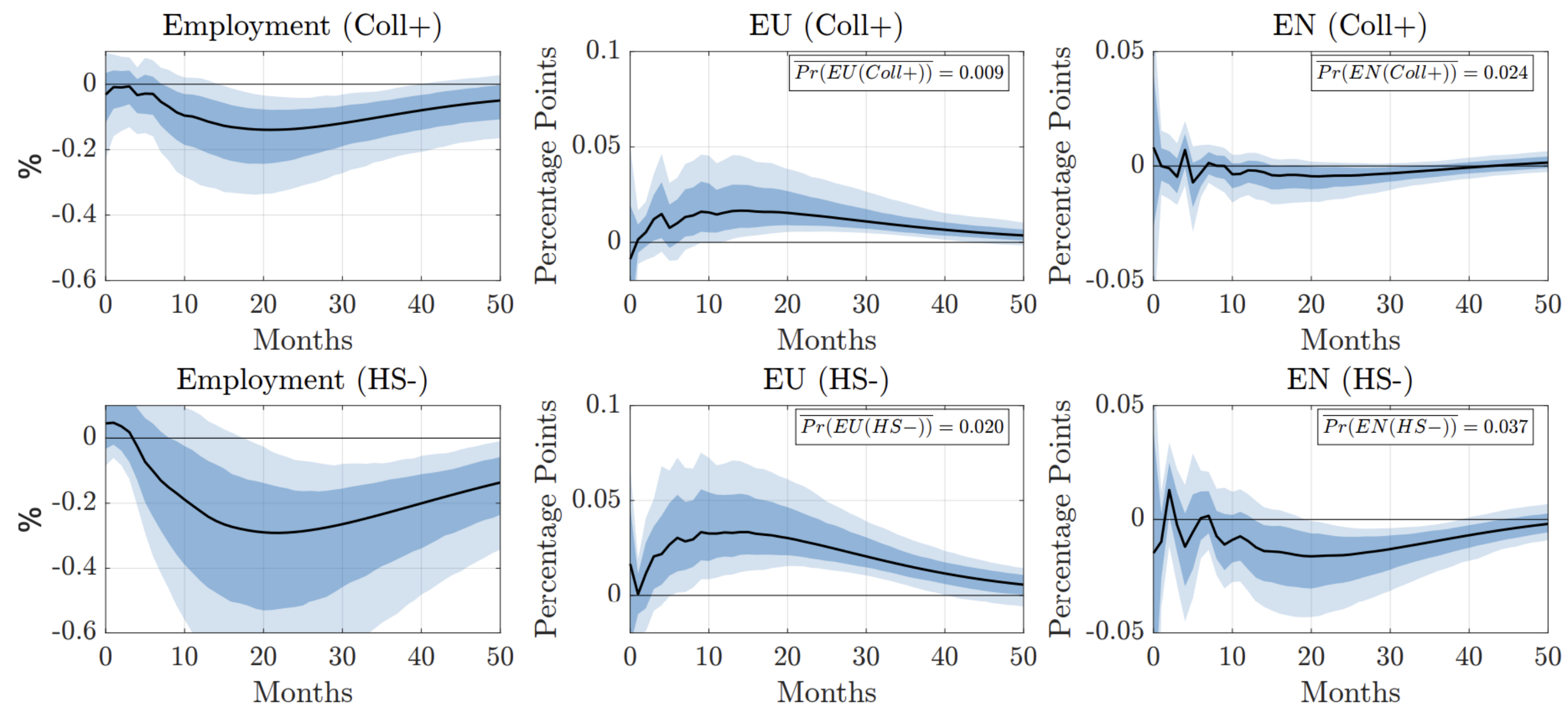


- Broadly Unchanged, but UN flow decreases by half
 - Change in the composition of U towards workers with higher labor force attachment
 - Reflects (only) 25% decline in the alleviating power of labor supply channel

HETEROGENOUS LABOR SUPPLY RESPONSE

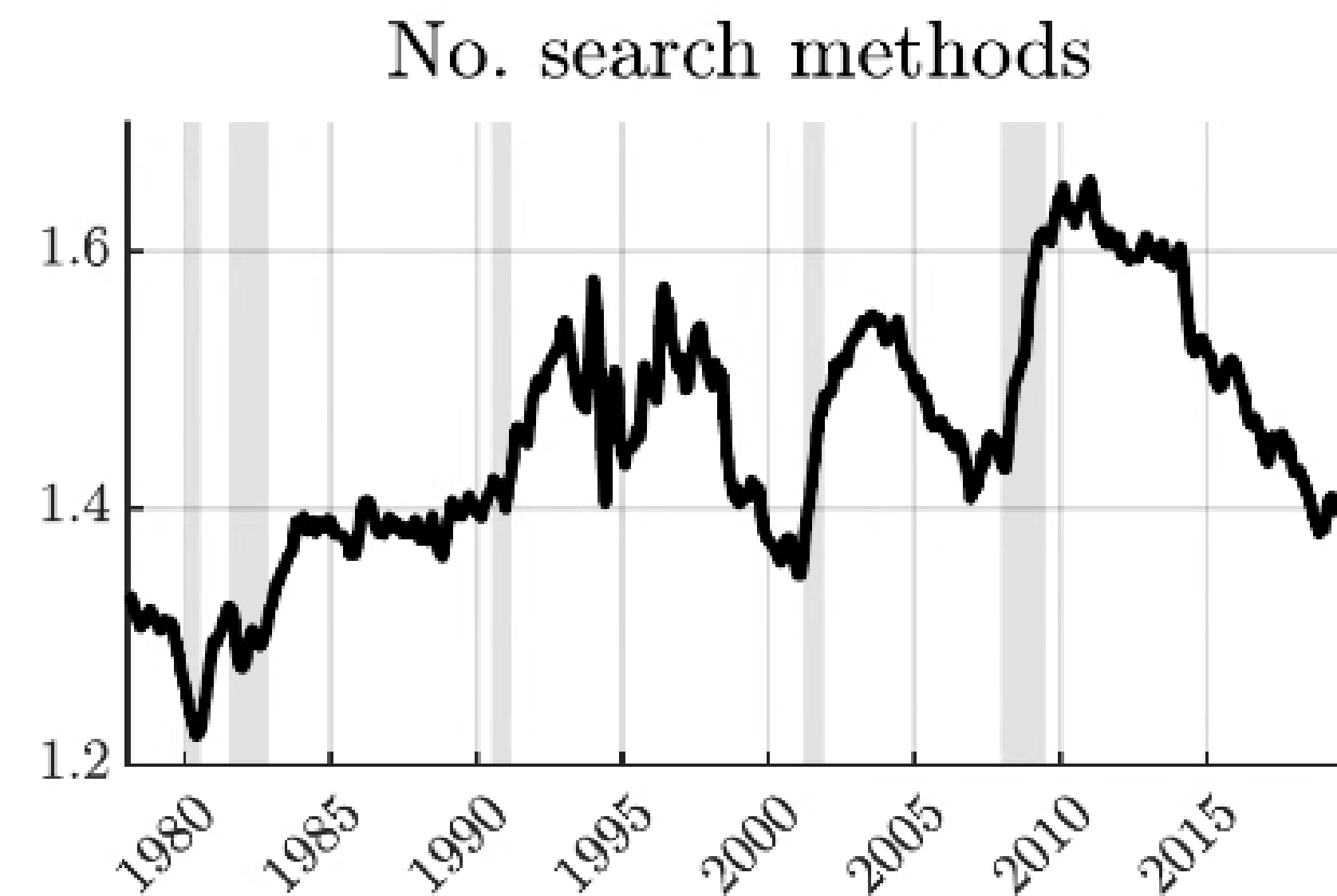
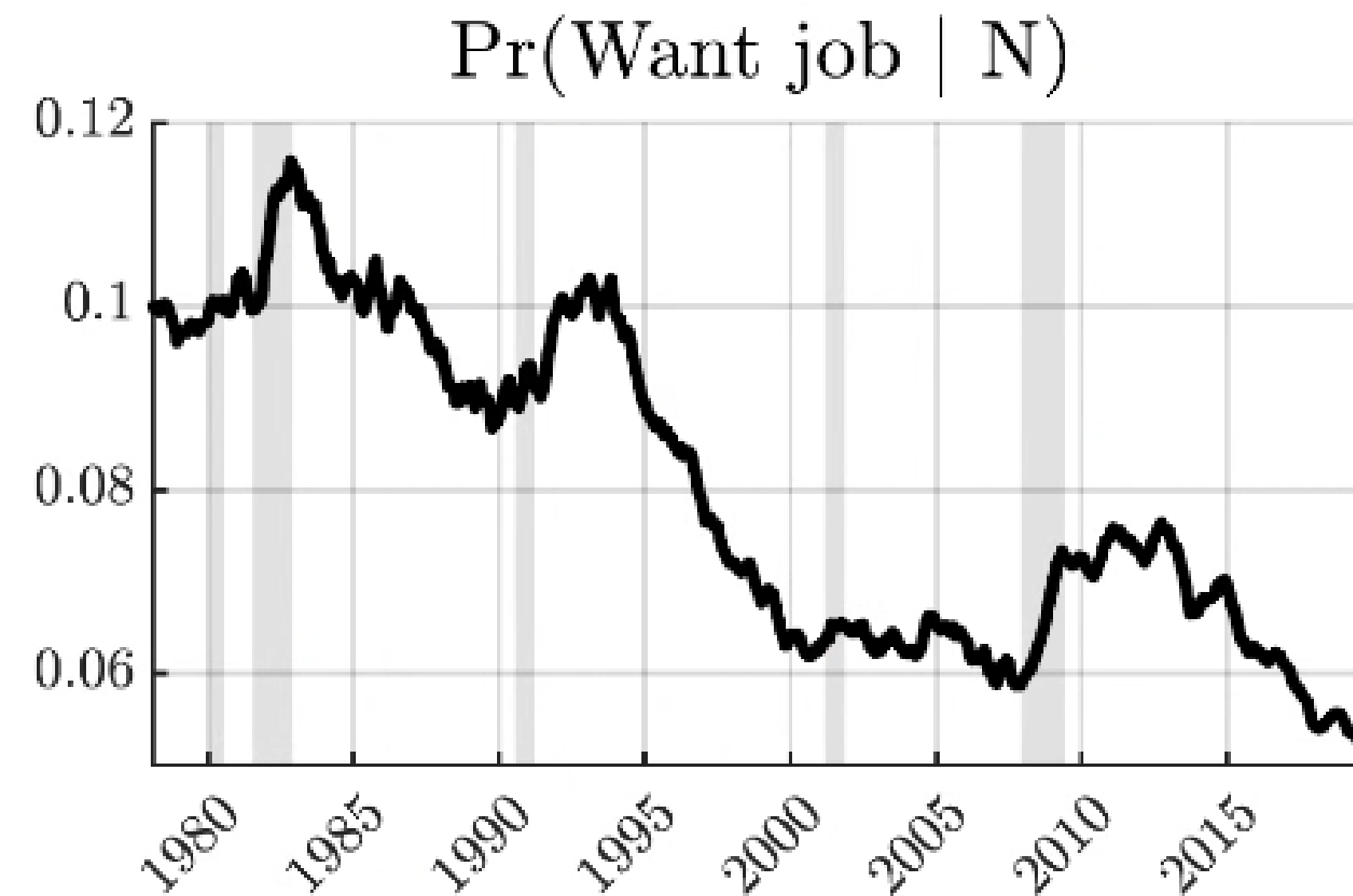
- Conjecture: The labor supply responses stem from Income Effect
 - Must observe people with fewer assets respond more
- As a proxy, examine responses by education
 - Assumption: Lower education group has lower asset

Figure: Responses by Education



INTENSIVE MARGIN OF LABOR SUPPLY

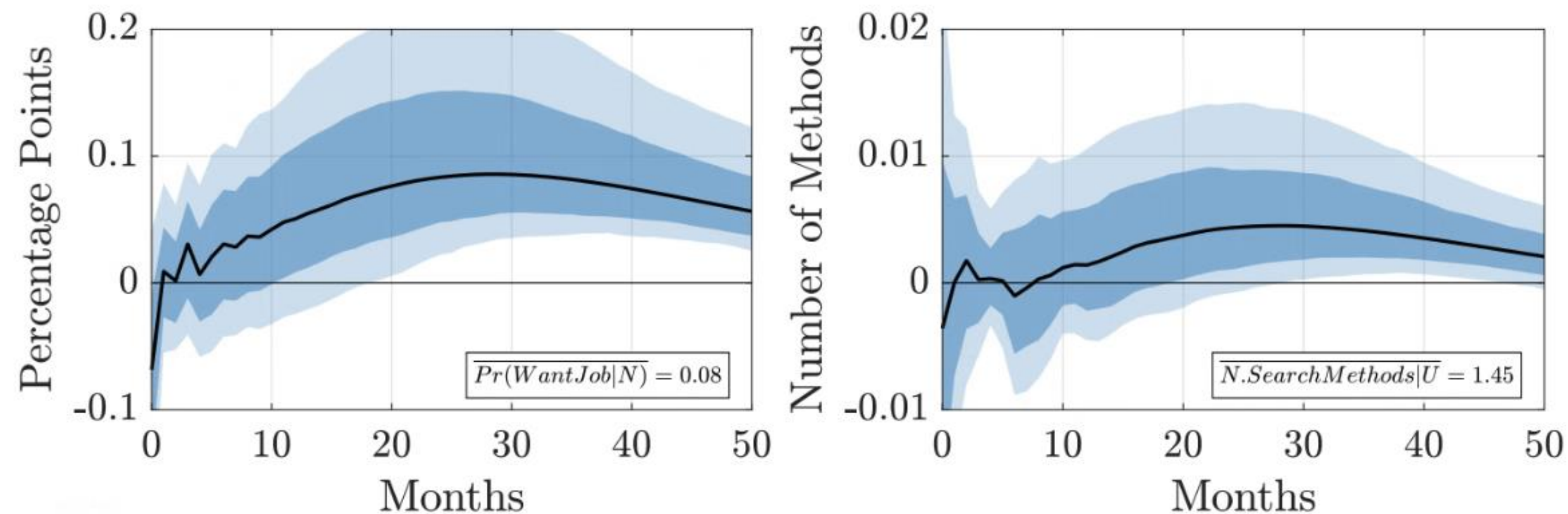
- This paper mostly about extensive margin of labor supply
- But intensive margin (search intensity) also counter-cyclical
 - 1) In recession, number of nonparticipants who wish to work increase
 - 2) Number of search methods increase (ATUS data. Mukoyama, Patterson and Sahin (2018))



INTENSIVE MARGIN OF LABOR SUPPLY

- Intensive Margin of Labor Supply also Increases:
- Measured by (1) % of Nonparticipants wanting job, (2) Search Intensity

FIGURE 4. Response of Intensive Margins of Labor Supply



THE MODEL

- **Objective:** Show labor supply decision matters even with sticky wage!
- Partial eq. (fixed wage) search model
 - Consumption fully insured within family
 - State b : Value of leisure, draw new b' at rate χ from dist. F

$$rV_0(b) = \max_{s \in \{0,1\}} \left\{ \frac{b - \psi \cdot \mathbb{I}\{s = 1\}}{\mu} + \left(\alpha s + (1 - \alpha) \right) \lambda \left[\max\{V_1(b), V_0(b)\} - V_0(b) \right] + \chi \left[\int_{\underline{b}}^{\bar{b}} V_0(b') dF(b') - V_0(b) \right] \right\}$$

Eq 1: Consumption Equivalent Value of Non-employment

THE MODEL

- State b : Value of leisure, draw new b' at rate χ from dist. F
- Choice : Passive Search ($s = 0$), Active Search ($s = 1$)
- s : active search incurs disutility, but higher job finding rate

Two Decisions:

- 1) Whether to search actively (s)
- 2) Accept or not accept an offer

$$\begin{aligned} rV_0(b) = \max_{s \in \{0,1\}} & \left\{ \frac{b - \psi \cdot \mathbb{I}\{s = 1\}}{\mu} \right. \\ & + \left(\alpha s + (1 - \alpha) \right) \lambda \left[\max\{V_1(b), V_0(b)\} - V_0(b) \right] \\ & \left. + \chi \left[\int_{\underline{b}}^{\bar{b}} V_0(b') dF(b') - V_0(b) \right] \right\} \end{aligned}$$

Eq 1: Consumption Equivalent Value of Non-employment

THE MODEL

- Employee's Value Function
- Decision: Whether to stay or quit after observing the leisure $\mathbf{b}$ draw

$$rV_1(b) = \max \left\{ rV_0(b), w + \delta[V_0(b) - V_1(b)] + \chi \left[\int_{\underline{b}}^{\bar{b}} \max\{V_0(b'), V_1(b')\} dF(b') - V_1(b) \right] \right\}$$

Eq 2: Consumption Equivalent Value of Employees

THE MODEL: CUTOFFS

- Characterization: $V_1(b) - V_0(b)$ decreasing in b . Two cutoffs arise:

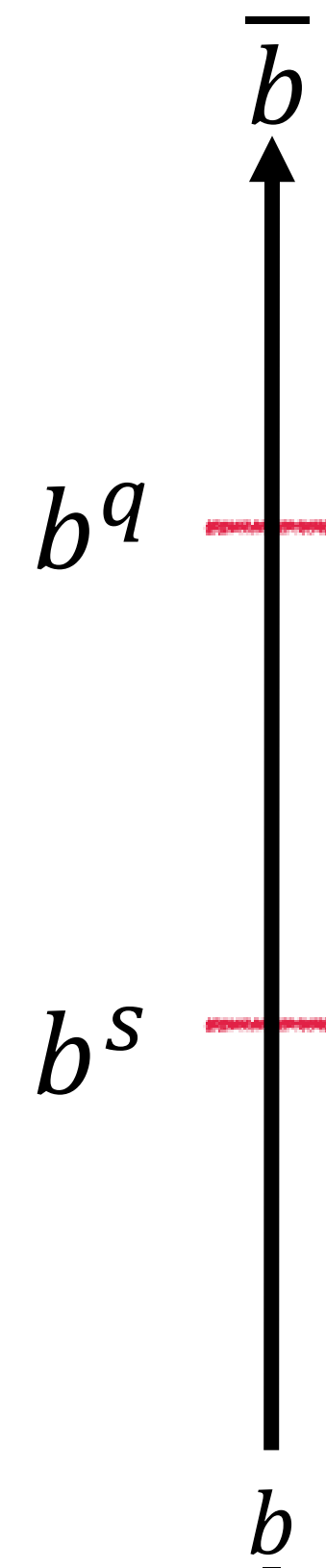
3. Non-employed not accept any offer

Indiff. between working & not-working

2. Non-employed search passively,
but accepts if offered

Indiff. between active & passive search

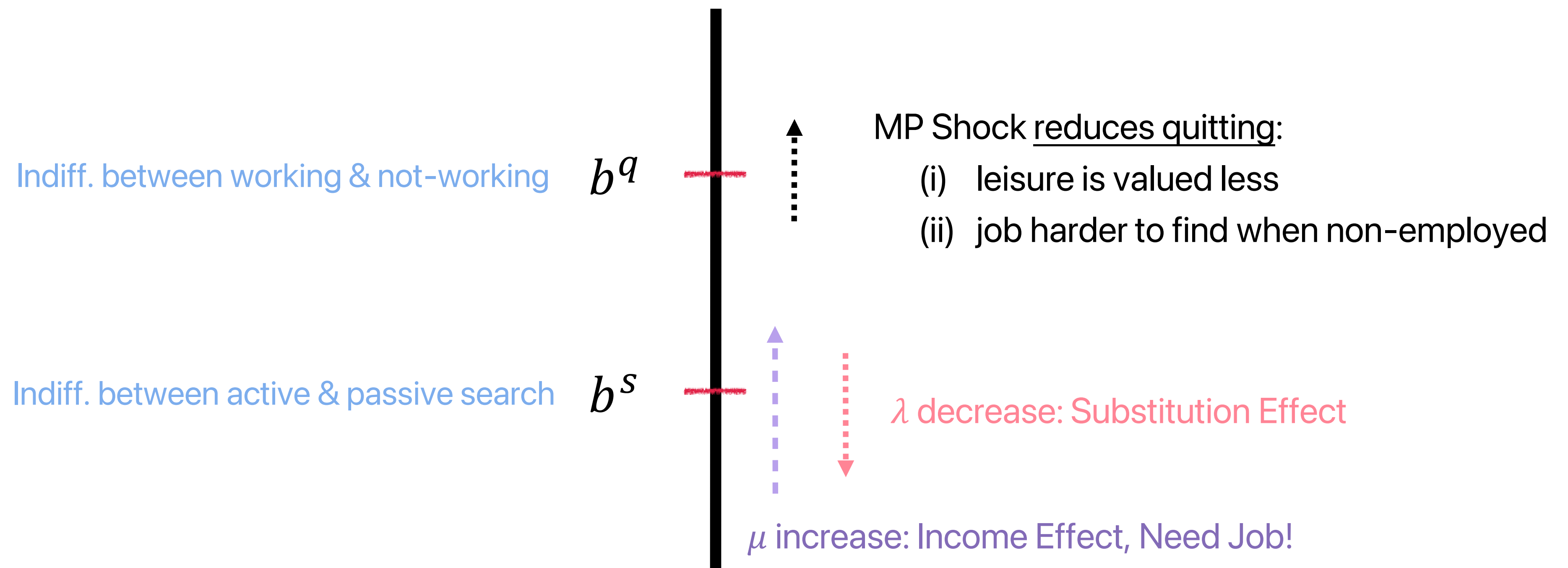
1. Non-employed prefer active search



THE MODEL: CUTOFFS

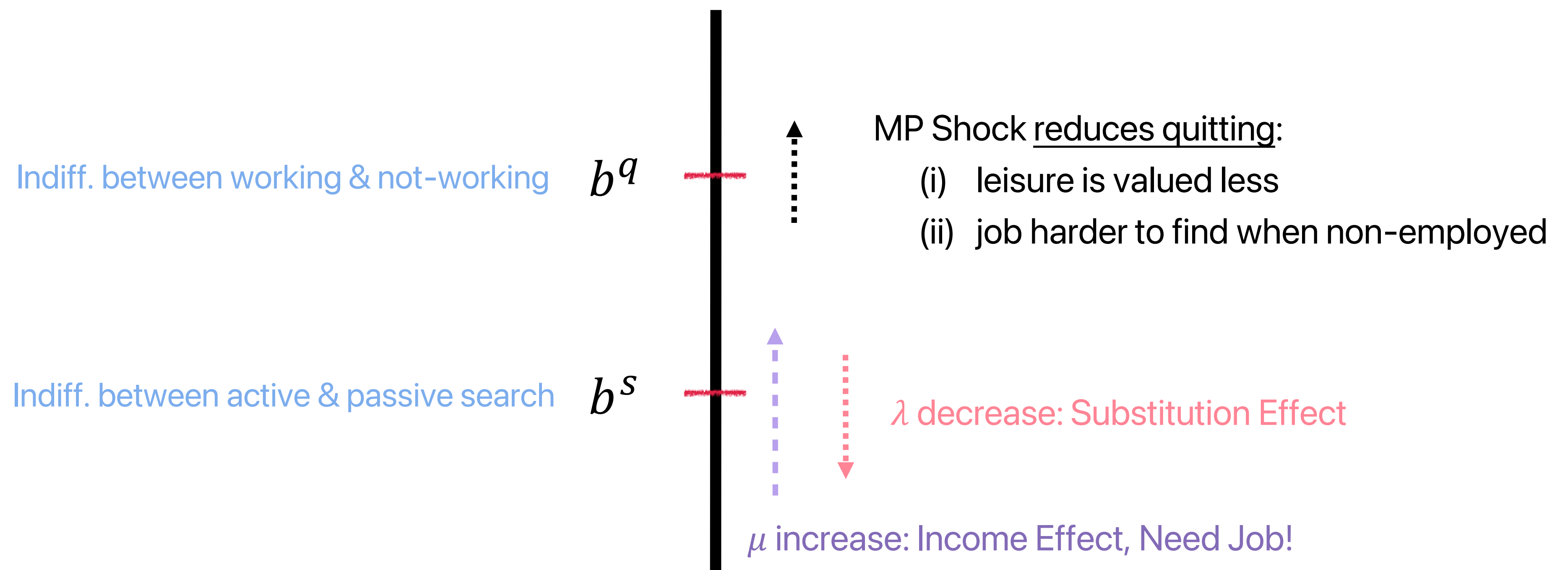
○ Monetary Policy Shock Modeled as:

- 1) Increase in Marginal Utility of Consumption μ : Jobs more valuable (Income Effect)
- 2) Reduction in job finding rate λ : Less incentives to search (Substitution Effect)



THE MODEL: CUTOFFS

- Empirical evidence of greater search behavior implies : Income Effect > Substitution Effect
- Implications for Sticky Wage NK Model : Not innocuous to preclude labor supply response



CONCLUSION

- Labor supply response to contractionary MP Shock is Large
 - Channel 1) Increased job-seeking behavior
 - Channel 2) Less quits to non-employment
 - ∴ Mitigate decline in labor demand from MP tightening by half
- Sticky wage model to show that even w/ sticky wage, labor supply decision can matter
- Future Research: Incorporating such features to full NK model (e.g.: Alves and Violante, 2023)

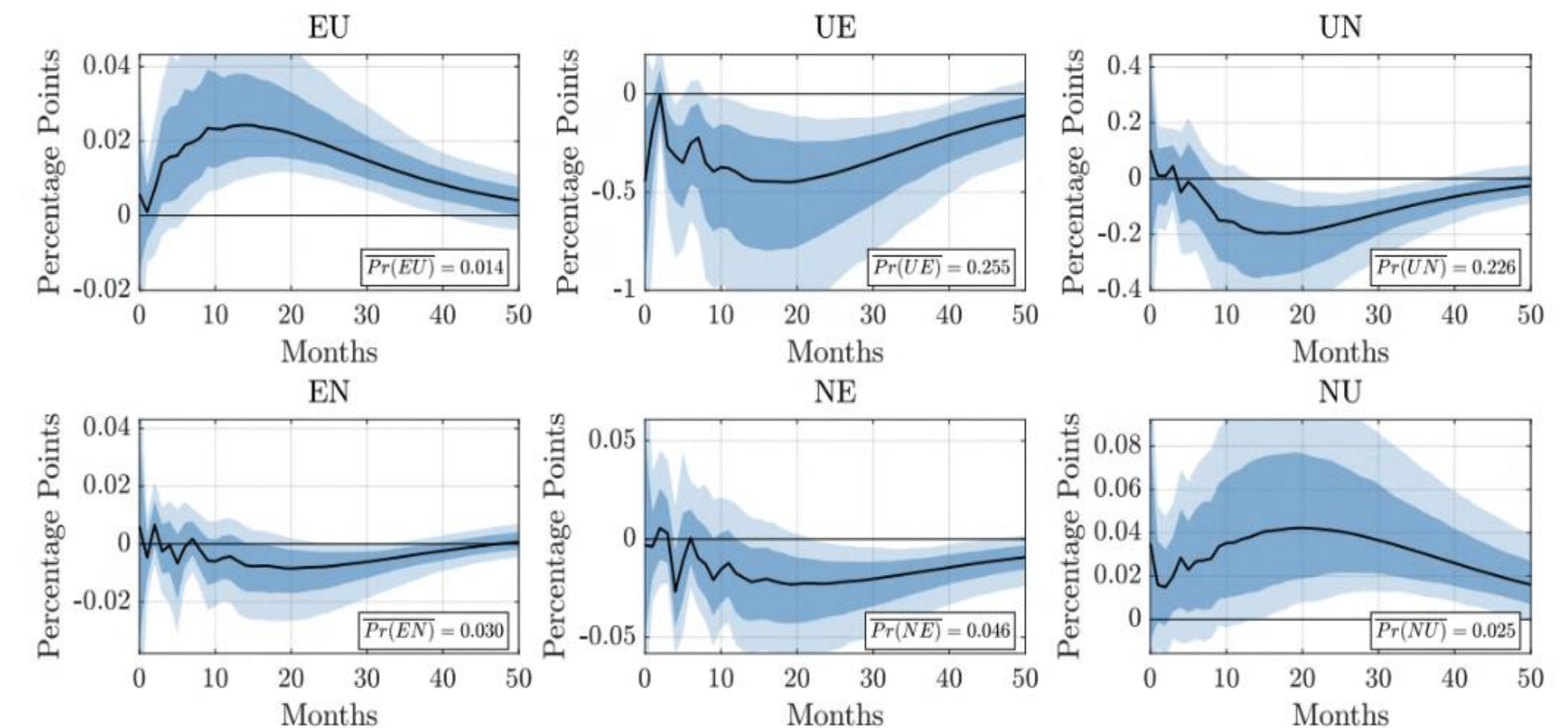
APPENDIX: RESPONSE OF OTHER VARIABLES

- Job-to-Job transition rates no significant response
- Posting of vacancies decrease
- Individual level wage growth responds very little (suggests wage stickiness)

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APPENDIX: CONSTRUCTING HYPOTHETICAL IRF

$$\begin{bmatrix} E \\ U \\ N \end{bmatrix}_{t+1} = \begin{bmatrix} 1 - p_{EU} - p_{EN} & p_{UE} & p_{NE} \\ p_{EU} & 1 - p_{UE} - p_{UN} & p_{NU} \\ p_{EN} & p_{UN} & 1 - p_{NE} - p_{NU} \end{bmatrix}_{t+1} \begin{bmatrix} E \\ U \\ N \end{bmatrix}_t.$$



1. Assume a given flow remains at its average level. For others, take the numbers of estimated IRFs
2. Feed IRFs into the Markov Chain.
3. Use the implied $\{E, U, N\}_t$ to calculate the unemployment rate

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APPENDIX : BAUER AND SWANSON (AER, 2023)

$$BCrev_t = \alpha + \theta mps_t + \varepsilon_t$$

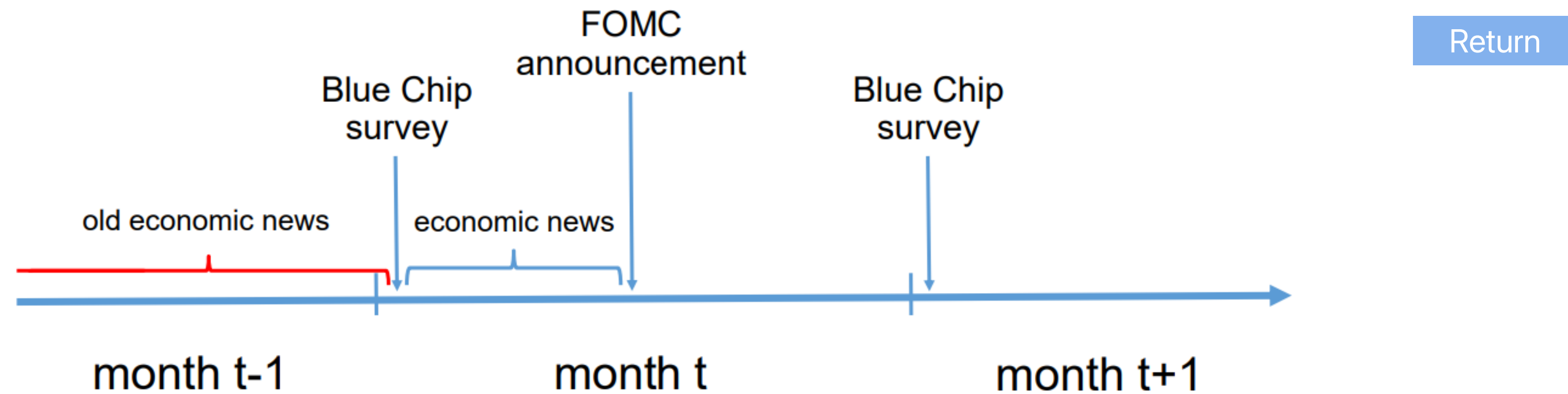
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1-month change in Blue Chip Forecast
around FOMC announcement

FOMC announcement surprise
in 30-min window

- t indexes FOMC announcements
- Empirical work sometimes : $\theta > 0$. (Standard macro model predicts opposite sign for GDP)
 - The “Fed Information Effect” Story (Nakamura-Steinsson 2018)
 - Fed better forecaster; when Fed lowers rate, indicative of worse economy so lower GDP forecast

APPENDIX : BAUER AND SWANSON (AER, 2023)



- Challenges this w/ "Fed info response channel": Fed and forecasters respond to incoming news
 - Shows economic news (macro, financial news) predicts both BC Revs and MPS
 - MPS correlated ex post with economic news, violating IV exogeneity condition
 - Construct new measures with variations around Fed Chair speeches & orthogonalized wrt news

APPENDIX : PROXY SVARs IDENTIFICATION SCHEME

Return

- If there exists instrument that satisfies

$$\mathbb{E} [\varepsilon_t^{\text{Demand}} \mathbf{z}_t'] = \mathbf{0},$$

$$\mathbb{E} [\varepsilon_t^{\text{MonPol}} \mathbf{z}_t'] = \mathbf{c}$$

- Then, We can identify one column of the reduced-form shock matrix:

$$\begin{bmatrix} y_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} - & b_{12} \\ - & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{\text{Demand}} \\ \varepsilon_t^{\text{MonPol}} \end{bmatrix}$$

APPENDIX : PROXY SVARs IDENTIFICATION SCHEME

[Return](#)

- Reduced form residuals are linear combinations of two structural shocks:

$$\begin{cases} u_{yt} = b_{11}\varepsilon_t^{\text{Demand}} + b_{12}\varepsilon_t^{\text{MonPol}} \\ u_{rt} = b_{21}\varepsilon_t^{\text{Demand}} + b_{22}\varepsilon_t^{\text{MonPol}} \end{cases}$$

- First-stage Regression: Estimates b_{22} by scaling factor

$$u_{rt} = \beta z_t + \xi_t$$

$$\beta = \frac{b_{22}c}{\text{Var}(z_t)} \qquad b_{22} = \beta \text{Var}(z_t) / c$$

APPENDIX : PROXY SVARs IDENTIFICATION SCHEME

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[Return](#)

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- Second-stage Regression: Estimates b_{12}' $u_{yt} = \gamma \hat{u}_{rt} + \zeta_t$

$$\gamma = \frac{b_{12}}{b_{22}}$$

* One can find all B matrix. (GK2015, Footnote 4)

- Normalize $b_{22} = 1$. And we have identified:

$$\begin{bmatrix} y_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} - & \gamma \\ - & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_t^{\text{Demand}} \\ \varepsilon_t^{\text{MonPol}} \end{bmatrix}$$